

Random Variables

Question 1. You are participating in a game where you have two chances to throw a ball toward a target hole. The rules of the game are as follows: If no balls land in the hole, you lose 10 Taka. If exactly one ball lands in the hole, you neither gain nor lose anything. If both balls land in the hole, you win 50 Taka. It is known that only 10% of the balls land in the holes, and the throws are independent.

- Construct a probability mass function (PMF) considering the gain or loss in the game as a random variable.
- Calculate the expected gain or loss, and determine if playing the game is favorable.
- Find the standard deviation of the gain or loss.

Solution:

- Let be X the random variable representing the gain or loss in Taka. We have three possible outcomes:

No balls land in the hole: Loss 10 Taka; and the probability:

$$P(X = -10) = {}^2C_0 \times (0.1)^0 \times (1 - 0.1)^2 = 0.81 \text{ (Using binomial distribution)}$$

Exactly one ball lands in the hole: No gain or loss (0 Taka); and the probability:

$$P(X = 0) = {}^2C_1 \times (0.1)^1 \times (1 - 0.1)^1 = 0.18$$

Both balls land in the hole: Win 50 Taka; and the probability:

$$P(X = 50) = {}^2C_2 \times (0.1)^2 \times (1 - 0.1)^0 = 0.01$$

Therefore, the probability mass function (PMF) is:

$X = x$	-10	0	50
$p(x)$	0.81	0.18	0.01

- Expected Gain or Loss:

The expected value ($E(X)$) is calculated as:

$$E(X) = \sum x \cdot p(x)$$

$$= (-10) \times 0.81 + 0 \times 0.18 + 50 \times 0.01$$

$$= -8.1 + 0 + 0.5 = -7.6$$

So, the expected loss is 7.6 Taka. Since the expected value is negative, playing the game is not favorable.

- Standard Deviation:

First, calculate the variance ($Var(X)$):

$$Var(X) = E(X^2) - [E(X)]^2$$

Where:

$$\begin{aligned}
 E(X^2) &= \sum x^2 \cdot p(x) \\
 &= (-10)^2 \times 0.81 + 0^2 \times 0.18 + 50^2 \times 0.01 \\
 &= 81 + 0 + 25 = 106 \\
 \therefore \text{Var}(X) &= 106 - (-7.6)^2 \\
 &= 106 - 57.76 = 48.24
 \end{aligned}$$

Finally, the standard deviation ($\sigma(X)$) is:

$$\sigma(X) = \sqrt{48.24} \approx 6.95$$

Thus, the standard deviation of the gain or loss is approximately 6.95 Taka.

Question 2. Suppose that 2 batteries are randomly chosen from a box containing 10 batteries of which 7 are good and 3 are defective. Let X denote the number of defective batteries chose.

- Summarize the probability function of X .
- Calculate the probability of
 - Exactly one defective battery chose
 - More than or equal to one defective battery chose
- Determine the mean and standard deviation of this probability function.

Solution:

- Here, X denote the number of defective batteries chose which values are 0, 1, and 2.

Probability of 0 defective battery, 2 good batteries: $P(0) = \frac{{}^3C_0 \times {}^7C_2}{{}^{10}C_2} = \frac{24}{45}$

Probability of 1 defective battery, 1 good battery: $P(1) = \frac{{}^3C_1 \times {}^7C_1}{{}^{10}C_2} = \frac{42}{45}$

Probability of 2 defective battery, 0 good battery: $P(2) = \frac{{}^3C_2 \times {}^7C_0}{{}^{10}C_2} = \frac{21}{45}$

Thus, the probability function of X

X	0	1	2
$P(x)$	$\frac{24}{45}$	$\frac{42}{45}$	$\frac{21}{45}$

- $P(\text{Exactly one defective battery chose}) = P(1) = \frac{42}{45}$

$P(X \geq 1) = P(1) + P(2) = ???$

- Mean, $E(X) = (0 \times \frac{24}{45}) + (1 \times \frac{42}{45}) + (2 \times \frac{21}{45}) = ???$

Comment:

Variance, $V(X) = E(X^2) - \{E(X)\}^2 = ???$

Comment:

Question 3. A random variable X has the following probability distribution:

$$P(X = k) = \begin{cases} \frac{k}{10}; & \text{for } k = 1, 2, 3, 4 \\ 0; & \text{Otherwise} \end{cases}$$

- Verify if this is a valid probability distribution.
- Find $P(X > 1)$ and $P(X < 4)$
- Calculate the expected value $E(X)$. Calculate $E(X^2)$.
- Calculate the variance $\text{Var}(X)$

Solution:

- Verify if this is a valid probability distribution:

$$P(1) + P(2) + P(3) + P(4) = \frac{1}{10} + \frac{2}{10} + \frac{3}{10} + \frac{4}{10} = 1 \text{ (valid)}$$

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$$P(X > 1) = 1 - P(X = 0) - P(X = 1) = 1 - \frac{0}{10} - \frac{1}{10} = \frac{9}{10}$$

$$P(X < 4) = 1 - P(X = 4) = 1 - \frac{4}{10} = \frac{6}{10}$$

- Calculate the expected value $E(X)$:

$$E(X) = \sum kP(X = k) = 1 \cdot \frac{1}{10} + 2 \cdot \frac{2}{10} + 3 \cdot \frac{3}{10} + 4 \cdot \frac{4}{10} = \frac{1 + 4 + 9 + 16}{10} = \frac{30}{10} = 3$$

Calculate $E(X^2)$:

$$E(X^2) = 1^2 \cdot \frac{1}{10} + 2^2 \cdot \frac{2}{10} + 3^2 \cdot \frac{3}{10} + 4^2 \cdot \frac{4}{10} = \frac{1 + 8 + 27 + 64}{10} = \frac{100}{10} = 10$$

- Calculate the variance $\text{Var}(X)$:

$$\text{Var}(X) = E(X^2) - (E(X))^2 = 10 - 3^2 = 10 - 9 = 1$$

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Question 4. The probability function of a discrete random variable X is

$$P(X = x) = \begin{cases} \alpha \left(\frac{3}{4}\right)^x & ; x = 0, 1, 2, 3 \\ 0 & ; \text{Elsewhere} \end{cases}$$

- What is the value of α ?
- Find the value of $P(X \leq 3)$.
- Calculate the value of $E(X)$ and $E(X^2)$.

Solution:

- Since $f(x)$ is a probability function, $\sum_x P(x) = 1$

But, $P(0) = \alpha \left(\frac{3}{4}\right)^0 = \alpha$, $P(1) = \alpha \left(\frac{3}{4}\right)^1 = \alpha \left(\frac{3}{4}\right)$, $P(2) = \alpha \left(\frac{3}{4}\right)^2 = \alpha \left(\frac{9}{16}\right)$; and $P(3) = \alpha \left(\frac{3}{4}\right)^3$ and so on. Hence,

$$1 = \alpha + \alpha \left(\frac{3}{4}\right) + \alpha \left(\frac{3}{4}\right)^2 + \alpha \left(\frac{3}{4}\right)^3$$

$$\Rightarrow 1 = \alpha \left(1 + \frac{3}{4} + \left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^3 \right)$$

$$\Rightarrow 1 = 2.73\alpha$$

$$\Rightarrow \alpha = 0.37$$

- Hence the complete pmf of X is $P(x) = 0.37 \left(\frac{3}{4}\right)^x$, $x = 0, 1, 2, 3$

$$\text{Also, } P(X \leq 2) = 0.37 + 0.37 \left(\frac{3}{4}\right) + 0.37 \left(\frac{9}{16}\right) = 0.84$$

$$E(X) = \sum xP(X = x) = 0(P(X = 0)) + 1(P(X = 1)) + 2(P(X = 2)) + 3(P(X = 3))$$

$$= 0 + \frac{3}{4} + 2 \left(\frac{3}{4}\right)^2 + 3 \left(\frac{3}{4}\right)^3 = ???$$

$$E(X^2) = \sum x^2 P(X = x) = ???$$

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Question 5. At BRAC University, students enrolled in the STA 201 course are required to submit 4 assignments throughout the semester. Let X be the discrete random variable representing the number of assignments submitted on time by a student out of the 4 total assignments. The probability mass function (PMF) of X is given by:

$$P(X = x) = \begin{cases} 0.05; & \text{if } x = 0 \\ 0.20; & \text{if } x = 1 \\ 0.35; & \text{if } x = 2 \\ 0.35; & \text{if } x = 3 \\ 0.05; & \text{if } x = 4 \end{cases}$$

If a student is randomly selected, then:

- Determine the probability that exactly 3 assignments are submitted on time.
- Compute the expectation (mean) of the random variable X . Based on the computed mean, can you determine whether a randomly selected student is likely to get full marks in assignments, given that full marks are awarded if at least 3, $(n - 1)$, assignments are submitted on time?
- Compute the variance of the random variable X .
- Suppose your instructor counts an additional assignment for every assignment submitted (e.g., if you submit 2 assignments, it will be counted as 3). Define a new random variable Y to represent this adjusted number of assignments. Determine the expectation and variance of Y . Based on the new mean, can you determine whether a randomly selected student is likely to get full marks in assignments? Also, find the probability mass function (PMF) of Y .

Solution:

- To determine the probability that exactly 3 assignments are submitted on time, we use the PMF provided:

$$P(X = 3) = 0.35$$

- To compute the expectation (mean) of the random variable X , we use the formula for the expected value of a discrete random variable:

$$E(X) = \sum_{x=0}^4 x P(X = x)$$

Substituting the given PMF values:

$$E(X) = (0 \cdot 0.05) + (1 \cdot 0.20) + (2 \cdot 0.35) + (3 \cdot 0.35) + (4 \cdot 0.05)$$

$$E(X) = 0 + 0.20 + 0.70 + 1.05 + 0.20$$

$$E(X) = 2.15$$

A student is likely to get full marks if they submit at least 3 assignments on time. Given that $E(X) = 2.15$, which is less than 3, it is not very likely that a randomly selected student will get full marks.

- c. To compute the variance of X , we first need $E(X^2)$:

$$E(X^2) = \sum_{x=0}^4 x^2 P(X = x)$$

$$E(X^2) = (0^2 \cdot 0.05) + (1^2 \cdot 0.20) + (2^2 \cdot 0.35) + (3^2 \cdot 0.35) + (4^2 \cdot 0.05)$$

$$E(X^2) = 0 + 0.20 + 4 \cdot 0.35 + 9 \cdot 0.35 + 16 \cdot 0.05$$

$$E(X^2) = 0.20 + 1.40 + 3.15 + 0.80$$

$$E(X^2) = 5.55$$

The variance is given by:

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

$$\text{Var}(X) = 5.55 - (2.15)^2$$

$$\text{Var}(X) = 5.55 - 4.6225$$

$$\text{Var}(X) = 0.9275$$

- d. Suppose the instructor counts an additional assignment for every assignment submitted, i.e., if a student submits X assignments, it is counted as $Y = X + 1$. Then:

$$E(Y) = E(X + 1) = E(X) + 1$$

$$E(Y) = 2.15 + 1 = 3.15$$

With the new mean $E(Y) = 3.15$, a randomly selected student is likely to get full marks in assignments if the new mean number of assignments (including the additional count) is at least 3. Since $E(Y)$ is 3.15, which is greater than 3, on average, a randomly selected student will get full marks under the new counting scheme.

The variance of Y is the same as the variance of X because adding a constant does not affect the variance:

$$\text{Var}(Y) = \text{Var}(X) = 0.9275$$

The PMF of Y can be derived from the PMF of X by shifting the values by 1:

$$P(Y = y) = P(X = y - 1)$$

Thus, the PMF of Y is:

$$P(Y = y) = \begin{cases} 0.05; & \text{if } y = 1 \\ 0.20; & \text{if } y = 2 \\ 0.35; & \text{if } y = 3 \\ 0.35; & \text{if } y = 4 \\ 0.05; & \text{if } y = 5 \end{cases}$$

Question 6. Suppose in a game, a participant gets two chances to draw a token each time from a gamble box. The participant receives prizes according to the token they draw. There are 5 tokens for special prizes and 8 tokens for ordinary prizes. Let X denote the number of tokens for special prizes chose.

- Summarize the probability function of X .
- Calculate the probability of exactly two special tokens chose
- Determine the mean and standard deviation of this probability function.

Solution:

- Here, X denote the number of tokens for special prizes chose which values are 0, 1, and 2.

Probability of 0 special token, 2 ordinary token: $P(0) = \frac{{}^5C_0 \times {}^8C_2}{{}^{13}C_2} = \frac{28}{78}$

Probability of 1 special token, 1 ordinary token: $P(1) = \frac{{}^5C_1 \times {}^8C_1}{{}^{13}C_2} = \frac{40}{78}$

Probability of 2 special token, 0 ordinary token: $P(2) = \frac{{}^5C_2 \times {}^8C_0}{{}^{13}C_2} = \frac{10}{78}$

Thus, the probability function of X

X	0	1	2
$P(X)$	$\frac{28}{78}$	$\frac{40}{78}$	$\frac{10}{78}$

- Exactly one special token chose,

$$P(\text{Exactly one special token chose}) = P(1) = \frac{40}{78}$$

- Mean, $E(X) = ???$

Standard deviation,

Hints:

$$V(X) = E(X^2) - (E(X))^2$$

$$E(X^2) = 0^2 \cdot \frac{28}{78} + 1^2 \cdot \frac{40}{78} + 2^2 \cdot \frac{10}{78}$$

Question 7. Let X be a random variable with probability distribution

x	0	1	2	3
$p(x)$	1/3	1/2	0	1/6

Determine,

- $P(X > 1)$ [Ans: 1/6]
- $P(X \leq 2)$ [Ans: 5/6]
- $E(X)$ [Ans: 1]
- $E(X^2)$ [Ans: 2]
- $E[(X - 1)^2]$ [Ans: 1]
- $Var(X)$ [Ans: 1]

Question 8. Katrina Kaif's Kay Beauty brand estimates the net profit on a new skincare product. This versatile brand would like to launch this product and try to estimate its profit during the 1st year. If its net profit is \$5 million, it is 'popular'; if it is \$2 million, it is 'fairly popular' and a loss of \$2 million if it is 'flop'. The brand assigns the following probability to first year expectations for the product; popular: 0.18, fairly popular: 0.45, and flop: 0.37. What are the expectations and standard deviation of the first-year net profit for this new product?

- Construct a probability mass function (PMF) considering the 1st net profit of the required brand. Is it a valid probability distribution?
- Estimate the probability of net profit less than or equal to \$2 million.
- Determine the mean and standard deviation of net profit for this new product.
- Calculate $E(2X+3)$ and $Var(4X+7)$.

Solution: Let, X be the random variable that represents net profit during 1st year.

- Probability mass function (PMF) based on the above situation:

$X=x$	-2	2	5
$P(x)$	0.37	0.45	0.18

This PMF will be valid if it satisfies $\sum P(x_i) = 1$

Here, $P(X = -2) + P(X = 2) + P(X = 5) = 0.37 + 0.45 + 0.18 = 1$

Therefore, the above PMF is valid.

- $P(X \leq 2) = P(X = -2) + P(X = 2) = 0.37 + 0.45 = 0.82$
- $$E(X) = \sum xP(X = x) = (-2)P(X = -2) + (2)P(X = 2) + (5)P(X = 5)$$

$$= (-2 \times 0.37) + (2 \times 0.45) + (5 \times 0.18) = 1.06$$

$$E(X^2) = \sum x^2 P(X = x) = (-2)^2 P(X = -2) + (2)^2 P(X = 2) + (5)^2 P(X = 5)$$

$$= (4 \times 0.37) + (4 \times 0.45) + (25 \times 0.18) = 7.78$$

$$Var(X) = E(X^2) - (E(X))^2 = 7.78 - (1.06)^2 = 6.6564$$

$$SD(X) = \sqrt{Var(X)} = \sqrt{6.66} = 2.58$$

$$d. E(2X + 3) = 2E(X) + 3 = 2 \times 1.06 + 3 = 5.12$$

$$Var(4X + 7) = 16Var(X) = 16 \times 6.6564 = 106.5024$$

Question 9. Bangladesh Bank has six tellers available to serve customers. The number of tellers busy with customers at peak time, say, 11:00 a.m. varies from day to day. So, it is a random variable denoted by X. It is known from the past records that the probability distribution of X is as follows.

Value of X	0	1	2	3	4	5	6
P(x)	0.03	0.05	0.08	0.15	K	0.26	0.22

- Find the value of K.
- Find the expected number of tellers busy with the customers at 11.00 a.m.
- Find the variance and coefficient of variation of the number of tellers busy with the customers at 11.00 a.m.
- Suppose the number of tellers busy with customers doubled and the resulting random variable is denoted by $Y = 2X$. Determine the variance and CV of Y and express it as a function of X.

Solution:

- To find K, we use the fact that the sum of all probabilities must equal 1:

$$0.03 + 0.05 + 0.08 + 0.15 + K + 0.26 + 0.22 = 1$$

$$K = 0.21$$

- The expected value $E(X)$ is calculated as follows:

$$E(x) = \sum_x x \cdot P(x) = 3.72$$

- The variance is calculated as

$$V(x) = E(X^2) - (E(X))^2 = 6.69$$

- The coefficient of variation is

$$CV = \frac{\sqrt{V(x)}}{E(x)} = 0.696$$

- Try yourself, the answer is: 26.79.